

MAT1002 Lecture 12,

Outline

- Ideal projectile
- Arc length parameter
- Tangent, normal and curvature

Ideal Projectile

We have derived that the positions (x, y) of an ideal projectile with horizontal launch angle α satisfy

$$x = (v_0 \cos \alpha)t, \quad y = (v_0 \sin \alpha)t - \frac{1}{2}gt^2, \quad t \geq 0.$$

Since $\frac{dy}{dt} = v_0 \sin \alpha - gt$, $\frac{dy}{dt} = 0$ at $t_0 \Leftrightarrow t_0 = \frac{v_0 \sin \alpha}{g}$.

By physical nature, y has a maximum at this t_0 , and
can also be seen by second derivative test

$$y_{\max} = v_0^2 \sin^2 \alpha / g - \frac{1}{2g} v_0^2 \sin^2 \alpha = \frac{v_0^2 \sin^2 \alpha}{2g}.$$

When object hits the ground again, $y=0$, so

$$(v_0 \sin \alpha)t - \frac{1}{2}gt^2 = 0 \Rightarrow v_0 \sin \alpha - \frac{1}{2}gt = 0 \Rightarrow t = \frac{2v_0 \sin \alpha}{g}.$$

Time hitting the ground

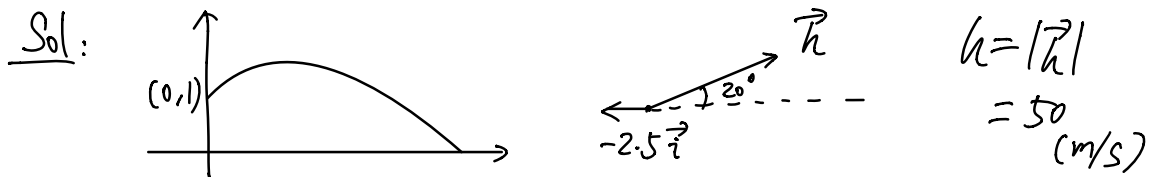
The total horizontal distance travelled by the object (when hitting the ground) is given by

$$v_0 \cos \alpha \left(\frac{2v_0 \sin \alpha}{g} \right) = \frac{v_0^2}{g} 2 \cos \alpha \sin \alpha = \frac{v_0^2}{g} \sin 2\alpha.$$

x-coordinate when hitting ground

EXAMPLE 5 ^{13.2.5} A baseball is hit when it is 1 m above the ground. It leaves the bat with initial speed of 50 m/s, making an angle of 20° with the horizontal. At the instant the ball is hit, an instantaneous gust of wind blows in the horizontal direction directly opposite the direction the ball is taking toward the outfield, adding a component of $-2.5\mathbf{i}$ (m/s) to the ball's initial velocity ($2.5 \text{ m/s} = 9 \text{ km/h}$).

- (a) Find a vector equation (position vector) for the path of the baseball.
 (b) How high does the baseball go, and when does it reach maximum height?
 (c) Assuming that the ball is not caught, find its range and flight time.



$$\begin{aligned}
 \text{(a)} \quad \vec{v}_0 &= \vec{h} - 2.5\vec{i} = (50 \cos 20^\circ - 2.5)\vec{i} + (50 \sin 20^\circ)\vec{j} \\
 \vec{a} &= -g\vec{j} \\
 \Rightarrow \vec{v}' &= \int \vec{v}'' dt = \int \vec{a} dt = -gt\vec{j} + \vec{v}'(0) = -gt\vec{j} + \vec{v}_0 \\
 \Rightarrow \vec{r} &= \int \vec{v}' dt = -\frac{1}{2}gt^2\vec{j} + t\vec{v}_0 + \vec{r}(0) \quad \langle 0,1 \rangle = \vec{j} \\
 &= -\frac{1}{2}gt^2\vec{j} + (50 \cos 20^\circ - 2.5)t\vec{i} + (50 \sin 20^\circ t)\vec{j} + \vec{j} \\
 &= \underbrace{(50 \cos 20^\circ - 2.5)t\vec{i}}_x + \underbrace{\left(-\frac{1}{2}gt^2 + 50 \sin 20^\circ t + 1\right)\vec{j}}_y
 \end{aligned}$$

(b) Max height: find t that satisfies $\frac{dy}{dt} = 0$. $\leftarrow$ Exercise.

(c) Flight time and range: find t that satisfies $y(t) = 0$. $\leftarrow$

Arc Length Parameter

The same curve can be parametrized in many different ways. One way is to use the so-called arc length parameter.

Arc Length

Definition

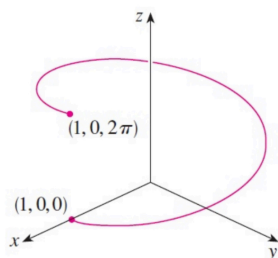
Let $\vec{r}: [a, b] \rightarrow \mathbb{R}^n$ be a **continuously differentiable** vector function (that is, $\vec{r}'$ is continuous). The **arc length** (or **length**) of a curve parametrized by $\vec{r}$ is defined to be the number L , where

$$L := \int_a^b |\vec{r}'(t)| dt,$$

assuming that the curve is traced exactly once (i.e. $\vec{r}$ is one-to-one). In particular, for a space curve ($n = 3$) parametrized by $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle$, its length is given by

$$\int_a^b \sqrt{\underbrace{(f'(t))^2}_{\left(\frac{dx}{dt}\right)^2} + \underbrace{(g'(t))^2}_{\left(\frac{dy}{dt}\right)^2} + \underbrace{(h'(t))^2}_{\left(\frac{dz}{dt}\right)^2}} dt.$$

e.g. The length of a part of the helix below is given by

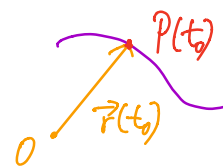


$$\vec{r}(t) = \langle \cos t, \sin t, t \rangle$$

$$L = \int_0^{2\pi} \sqrt{\sin^2 t + \cos^2 t + 1} dt = 2\pi\sqrt{2}.$$

The Arc Length Function

Consider a curve parametrized by a one-to-one continuously differentiable function $\vec{r}(t)$ defined for $t \in [a, b]$. Fix a value $t_0 \in [a, b]$, and define $s : [a, b] \rightarrow \mathbb{R}$ by



$$s(t) := \int_{t_0}^t |\vec{r}'(u)| du. \quad \leftarrow \text{"Signed distance from } P(t_0) \text{ to } P(t) \text{ along the curve"}$$

- $s(t) > 0$ if $t > t_0$;
- $s(t) < 0$ if $t < t_0$;
- By FTC, $s'(t) = \frac{ds}{dt} = |\vec{r}'(t)|$, the speed of the motion.
- If $\vec{r}'(t) \neq \vec{0} \quad \forall t$, then $s'(t) > 0 \quad \forall t \in I$, so $s \nearrow$ as $t \nearrow$.

Example

Re-parametrize the helix $\vec{r}(t) = \langle \cos(t), \sin(t), t \rangle$ with respect to arc length measured from $(1, 0, 0)$ in the direction of increasing t .

Sol: • $\vec{r}(t) = \langle 1, 0, 0 \rangle \Leftrightarrow t = 0$.

• $s(t) = \int_0^t |\vec{r}'(u)| du = \int_0^t \sqrt{2} du = \sqrt{2} t$.

• Take $s = s(t)$ as the new parameter (arc length parameter).

• $t = s/\sqrt{2}$.

• $\vec{r}(s/\sqrt{2}) = \langle \cos \frac{s}{\sqrt{2}}, \sin \frac{s}{\sqrt{2}}, \frac{s}{\sqrt{2}} \rangle =: \vec{\gamma}(s)$ is a new parametrization.

• New parameter domain: $s \in (-\infty, \infty)$.

Sometimes it is useful to parametrize a curve using the arc length parameter s , since arc length is a geometric concept and does not depend on motion, and considering it may reveal geometric properties of the curve (e.g. curvature) instead of motion properties (e.g. velocity).

Tangent, Normal and Curvature

Let $\vec{r} := \vec{r}(t)$ be the position vector of a curve, where $\vec{r}'$ is cts and $\vec{r}'(t) \neq \vec{0} \forall t$. Consider representing $\vec{r}$ using the arc length parameter s (base point $P(t_0)$ can be chosen in any way).

What is $d\vec{r}/ds$?

• Since s is $\nearrow$ as $t \nearrow$, $s = s(t)$ is one-to-one and thus is invertible. Since $s(t)$ is differentiable, dt/ds exists.

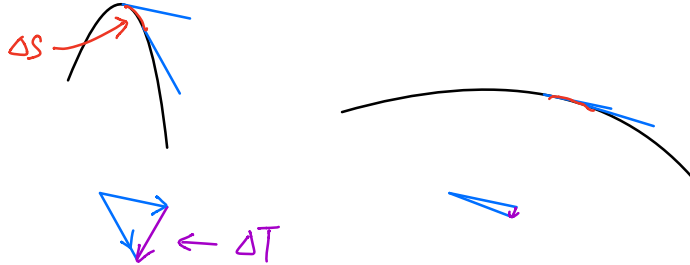
$$\bullet \frac{d\vec{r}}{ds} = \frac{d\vec{r}}{dt} \cdot \frac{dt}{ds} = \frac{\vec{r}'(t)}{|\vec{r}'(t)|} = \vec{T}(t) : \quad \boxed{\frac{d\vec{r}}{ds} = \vec{T}}$$

Unit tangent vector

Def: The **curvature** κ of a curve is "the rate at which its $\vec{T}$ turns per unit length along the curve". Formally,

$$\kappa := \left| \frac{d\vec{T}}{ds} \right|.$$

- The bigger κ is, the sharper turn the curve would make at a point.



- If position vector $\vec{r}$ is parametrized in t (which may not be s), and if the parametrization is smooth, then

$$\kappa = \left| \frac{d\vec{T}}{ds} \right| = \left| \frac{d\vec{T}/dt}{ds/dt} \right| = \frac{1}{|\vec{r}'(t)|} \left| \frac{d\vec{T}}{dt} \right|.$$

e.g. Consider three curves:

$$C_1: \vec{r}_1(t) = \langle t, t \rangle, \quad t \in \mathbb{R}.$$

$$C_2: \vec{r}_2(t) = \langle a \cos t, a \sin t \rangle, \quad t \in [0, 2\pi).$$

$$C_3: \vec{r}_3(t) = \langle t, t^2 \rangle, \quad t \in \mathbb{R}.$$

$$C_1: \vec{r}'_1(t) = \langle 1, 1 \rangle, \quad \vec{T} = \left\langle \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right\rangle, \quad \left| \frac{d\vec{T}}{dt} \right| = 0, \quad \kappa = 0$$

at all points. $(a > 0)$

$$C_2: \vec{r}'_2(t) = \langle -a \sin t, a \cos t \rangle, \quad \vec{T} = \frac{1}{a} \langle -\sin t, \cos t \rangle,$$

$$\frac{d\vec{T}}{dt} = \frac{1}{a} \langle -\cos t, -\sin t \rangle, \quad \left| \frac{d\vec{T}}{dt} \right| = \frac{1}{a}.$$

$$\therefore \kappa = \frac{1}{a} \text{ at all points.}$$

$$C_3: \vec{r}'_3(t) = \langle 1, 2t \rangle, \quad \vec{T} = \frac{1}{\sqrt{1+4t^2}} \langle 1, 2t \rangle,$$

$$\begin{aligned}
\frac{d\vec{T}}{dt} &= \left\langle -\frac{1}{2}(1+4t^2)^{-\frac{3}{2}} \cdot 8t, \frac{2\sqrt{1+4t^2} - 2t \cdot \frac{1}{2}(1+4t^2)^{-\frac{1}{2}} \cdot 8t}{1+4t^2} \right\rangle \\
&= \left\langle -4t(1+4t^2)^{-\frac{3}{2}}, (2(1+4t^2) - 8t^2)(1+4t^2)^{-\frac{3}{2}} \right\rangle \\
&= (1+4t^2)^{-\frac{3}{2}} \langle -4t, 2 \rangle
\end{aligned}$$

$$\Rightarrow \kappa(t) = \left| \frac{d\vec{T}}{dt} \right| = \frac{1}{(1+4t^2)^{3/2}} \sqrt{16t^2 + 4}.$$

Note that κ is maximized to 2 when $t=0$, and indeed the curve turns the most sharply at $(0,0)$.

(Exercise: plot $\kappa(t)$ using desmos.)

Principal Unit Normal

Consider the vector $\frac{d\vec{T}}{ds}$.

- It points to a direction in which $\vec{T}$ turns as the curve bends.
- Since $\vec{T}$ has length 1 for all points on curve, the terminal points of $\vec{T}$ all lie on the unit circle. By a result last time, if we consider the curve being parametrized by s , we see that $\frac{d\vec{T}}{ds}$ is orthogonal to $\vec{T}$ at every point.

Def: At a point (on a curve) where $\kappa \neq 0$, the **principal unit normal vector** is

$$\vec{N} := \frac{d\vec{T}/ds}{|d\vec{T}/ds|} = \frac{1}{\kappa} \frac{d\vec{T}}{ds}.$$

- If position vector $\vec{r}$ is parametrized in t (which may not s), and if the parametrization is smooth, then

$$\vec{N} = \frac{d\vec{r}/ds}{|d\vec{r}/ds|} = \frac{\frac{d\vec{r}}{dt} / \frac{ds}{dt}}{\left| \frac{d\vec{r}}{dt} / \frac{ds}{dt} \right|} = \frac{\frac{d\vec{r}}{dt} / \frac{ds}{dt}}{\left| \frac{d\vec{r}}{dt} \right| / \frac{ds}{dt}} = \frac{d\vec{r}/dt}{|d\vec{r}/dt|}.$$

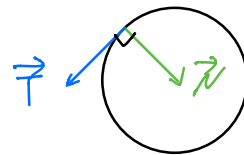
This formula allows the computation of $\vec{N}$ without finding s and K .

e.g. For the circle given by $\vec{r}(t) = \langle \cos 2t, \sin 2t \rangle$, $t \in [0, \pi)$,

$$\vec{r}'(t) = \langle -2\sin 2t, 2\cos 2t \rangle \Rightarrow \vec{T} = \langle -\sin 2t, \cos 2t \rangle$$

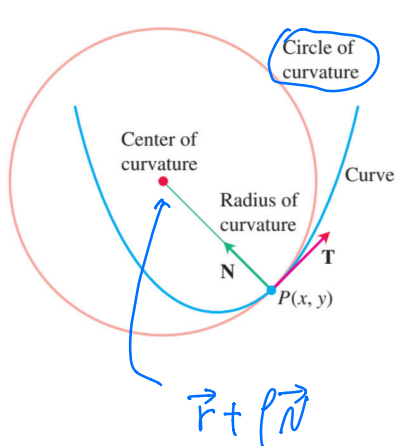
$$\Rightarrow \frac{d\vec{T}}{dt} = \langle -2\cos 2t, -2\sin 2t \rangle$$

$$\Rightarrow \vec{N} = \frac{d\vec{T}/dt}{|d\vec{T}/dt|} = \langle -\cos 2t, -\sin 2t \rangle$$



Def: Given a plane curve C and a point P on C at which $K \neq 0$, the **osculating circle** at P is the circle that

- shares the same tangent line at P as C , and;
- has the same curvature as C at P , and;
- has center that lies on the side to which $\vec{N}$ points.



= Osculating circle.

If ρ is the radius of curvature, then by previous example, we know $\kappa = \frac{1}{\rho}$, so $\rho = \frac{1}{\kappa}$.

The concept of κ and $\vec{N}$ can be defined for space curves, and can be computed using the same formulae.

e.g. For the helix given by $\vec{r}(t) = \langle a \cos t, a \sin t, bt \rangle$, ($a \geq 0, ab \neq 0$)

$$\vec{r}'(t) = \langle -a \sin t, a \cos t, b \rangle \Rightarrow \vec{T}(t) = \frac{1}{\sqrt{a^2 + b^2}} \langle -a \sin t, a \cos t, b \rangle$$

$$\Rightarrow \vec{T}'(t) = \frac{1}{\sqrt{a^2 + b^2}} \langle -a \cos t, -a \sin t, 0 \rangle$$

$$\Rightarrow \kappa = \frac{|\vec{T}'(t)|}{|\vec{T}'(t)|} = \frac{a}{\sqrt{a^2 + b^2}} \cdot \frac{1}{\sqrt{a^2 + b^2}} = \frac{a}{a^2 + b^2}$$

- $b \uparrow \Rightarrow \kappa \downarrow$
- $a = 0 \Rightarrow \kappa = 0$. (line)
- $b = 0 \Rightarrow \kappa = \frac{1}{a}$. (circle)